

REFLOW OVEN CONTROLLER DESIGN

Goals

- * $< 5^\circ\text{C}$ steady state error
- * $< 5^\circ\text{C}$ overshoot.

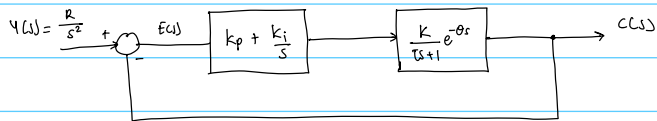
Open Loop Model of Reflow Oven

$$P(s) = \frac{K}{Ts+1} e^{-\theta s}$$

$\swarrow$ Gain $\swarrow$ Dead time
 $\nearrow$ Time constant

Type I System so Infinite steady state error with ramp input.

Going to use PI Controller to add a pole at the origin to make it a type I system.



$$T(s) = \frac{K(K_p s + K_i) e^{-\theta s}}{s(Ts+1) + K(K_p s + K_i) e^{-\theta s}}$$

$$E(s) = R(s) - C(s)$$

$$E(s) = R(s) - T(s)R(s)$$

$$E(s) = R(s) (1 - T(s))$$

$$= \frac{R}{s^2} \left(1 - \frac{(K_p s + K_i) e^{-\theta s}}{s(Ts+1) + K(K_p s + K_i) e^{-\theta s}} \right)$$

$$= \frac{R}{s^2} \left(\frac{s(Ts+1) + (K_p s + K_i) e^{-\theta s} - (K_p s + K_i) e^{-\theta s}}{s(Ts+1) + K(K_p s + K_i) e^{-\theta s}} \right)$$

$$E(s) = \frac{R}{s} \left(\frac{(Ts+1)}{s(Ts+1) + K(K_p s + K_i) e^{-\theta s}} \right)$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \frac{R}{K} \left(\frac{Ts+1}{s(Ts+1) + K(K_p s + K_i) e^{-\theta s}} \right)$$

$$e_{ss} = \frac{R}{K K_i}$$

Max ramp rate:

$$R = \frac{100-25}{90-0} = \frac{25}{18}$$

$$e_{ss} < 5$$

$$\frac{R}{K K_i} < 5$$

$$\frac{25}{18} < 5 K$$

$$K_i > \frac{5}{18 K}$$

Using SIMC Tuning



$$T(s) = \frac{P(s)C(s)}{1 + C(s)P(s)}$$

$$\Rightarrow \left(k_p + \frac{k_i}{s} \right) \left(\frac{1}{\tau_c s + 1} e^{-\theta s} \right)$$

$$\frac{k_p s + k_i}{s} \frac{1}{s(\tau_c s + 1)}$$

$$C(s) = \frac{1}{P(s)} \cdot \frac{T(s)}{1 - T(s)}$$

$\Rightarrow$

Desired transfer function: $T(s) = \frac{1}{\tau_c s + 1} e^{-\theta s}$

$$= \frac{1}{\frac{k e^{-\theta s}}{s\tau_c + 1}} \cdot \frac{1 \cdot e^{-\theta s}}{\tau_c s + 1 - e^{-\theta s}}$$

$$C(s) = \frac{\tau_c s + 1}{k} \left(\frac{1}{\tau_c s + 1 - e^{-\theta s}} \right)$$

$$\tau_c s + 1$$

$$e^{-\theta s} = 1 - \theta s$$

$$C(s) = \frac{\tau_c s + 1}{k(\tau_c s + 1 - (1 - \theta s))}$$

$$\frac{\tau_c s + 1}{k(\tau_c s + 1)}$$

$$= \frac{\tau_c s + 1}{k\tau_c s + k\theta s}$$

$$= \frac{\tau_c}{k(\tau_c + \theta)} + \frac{1}{k(\tau_c + \theta)} \cdot \frac{1}{s}$$

$$C(s) = k_p + k_i \frac{1}{s}$$

$$k_p = \frac{\tau_c}{k(\tau_c + \theta)} \quad k_i = \frac{1}{k(\tau_c + \theta)}$$

$$k_i = \frac{1}{k(\tau_c + \theta)}$$

$$k_i > \frac{5}{18k}$$

$$\frac{1}{k(\tau_c + \theta)} > \frac{5}{18k}$$

$$5(\tau_c + \theta) < 18$$

$$\tau_c + \theta < \frac{18}{5}$$

$$\tau_c < \frac{18}{5} - \theta$$

To meet overshoot spec, $\tau_c \gg \theta$ according to Dr Sigurd Stogestad

SYSTEM PARAMETERS

$$G(s) = \frac{667.2}{3955s + 1} e^{-29.2}$$

$$\tau_c = \frac{18}{5} - (29.2) < 0$$

PI CONTROLLER CAN'T TRACK REQUIRED RAMP ALONE. NEEDS FEEDFORWARD

FEED FORWARD DESIGN

$$r(t) = R t$$

$$u_{ff} = \frac{\tau R + r}{k}$$

$R \rightarrow$ Ramp rate $r \rightarrow$ Current temperature

$$\frac{k_p s + k_i}{s} \left(\frac{k}{\tau s + 1} \right)$$

$$\frac{(k_p s + k_i)(k)}{s(\tau s + 1) + (k_p s + k_i)k}$$

$$G(s) = \frac{(k_p s + k_i)k}{\tau s^2 + s + k_p k s + k_i k}$$

$$G(s) = \frac{k \left(s + \frac{k_i}{k_p} \right)}{s^2 + \frac{(k_p k + 1)}{\tau} s + \frac{k_i k}{\tau}}$$

$$s^2 + 2\xi\omega_n s + \omega_n^2$$

$$\omega_n^2 = \frac{k_i k}{\tau}$$

$$k_i = \frac{\omega_n^2 \tau}{k}$$

$$k_p = \frac{2\xi\omega_n \tau - 1}{k}$$